

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/321286678>

Decreasing sprint duration from 20 to 10 s during reduced-exertion high-intensity interval training (REHIT) attenuates the increase in maximal aerobic capacity but has no effect on...

Article in *Applied Physiology Nutrition and Metabolism* · November 2017

DOI: 10.1139/apnm-2017-0597

CITATIONS

2

READS

514

7 authors, including:



Gulbin Nalcakan

Ege University

23 PUBLICATIONS 29 CITATIONS

[SEE PROFILE](#)



Preeyaphorn Songsorn

University of Stirling

9 PUBLICATIONS 25 CITATIONS

[SEE PROFILE](#)



Ben Fitzpatrick

Ulster University

14 PUBLICATIONS 22 CITATIONS

[SEE PROFILE](#)



Yasin Yuzbasioglu

Ege University

2 PUBLICATIONS 2 CITATIONS

[SEE PROFILE](#)

Some of the authors of this publication are also working on these related projects:



Meta-Predict [View project](#)



Sport for LIFE: All Island [View project](#)



Decreasing sprint duration from 20 to 10 s during reduced-exertion high-intensity interval training (REHIT) attenuates the increase in maximal aerobic capacity but has no effect on affective and perceptual responses

Journal:	<i>Applied Physiology, Nutrition, and Metabolism</i>
Manuscript ID	apnm-2017-0597.R3
Manuscript Type:	Article
Date Submitted by the Author:	14-Nov-2017
Complete List of Authors:	Nalcakan, Gulbin Rudarli; Ege Univ, faculty of sport science Songsorn, Preeyaphorn; University of Stirling Fitzpatrick, Ben; University of Ulster Yüzbasıoglu, Yasin; Ege Univ, faculty of sport science Brick, Noel; University of Ulster Metcalf, Richard; Swansea University Vollaard, Niels; University of Stirling
Is the invited manuscript for consideration in a Special Issue? :	
Keyword:	VO2max, sprint interval training, SIT, Wingate sprint, affect

SCHOLARONE™
 Manuscripts

1 **Decreasing sprint duration from 20 to 10 s during reduced-exertion high-intensity interval training**
2 **(REHIT) attenuates the increase in maximal aerobic capacity but has no effect on affective and**
3 **perceptual responses**

4

5 Gulbin R Nalçakan¹, Preeyaphorn Songsorn², Ben L Fitzpatrick³, Yasin Yüzbasıoglu¹, Noel E Brick⁴,
6 Richard S Metcalfe^{3,5}, Niels BJ Vollaard^{2*}

7

8 ¹ Faculty of Sport Science, Ege University, 35040 İzmir, Turkey (gulbinrn@gmail.com,
9 yasinyuzbasiloglu@gmail.com)

10 ² Faculty of Health Sciences and Sport, University of Stirling, Stirling, FK9 4LA, UK
11 (preeyaphorn.songsorn1@stir.ac.uk, n.vollaard@stir.ac.uk)

12 ³ School of Sport, Ulster University, Derry/Londonderry, BT48 7JL, UK (bl.fitzpatrick@ulster.ac.uk)

13 ⁴ Psychology Research Institute, Ulster University, Derry/Londonderry, BT48 7JL, UK
14 (n.brick@ulster.ac.uk)

15 ⁵ Applied Sports Science Technology and Medicine Research Centre (A-STEM), Swansea University,
16 Swansea, SA1 8EN, UK (r.s.metcalfe@swansea.ac.uk)

17

18 *** Corresponding Author:**

19 Dr Niels Vollaard

20 Faculty of Health Sciences and Sport

21 University of Stirling

22 Stirling

23 FK9 4LA

24 UK

25 Phone: +44 1786 466488

26 Email: n.vollaard@stir.ac.uk

27 **ABSTRACT**

28

29 **Purpose:** Recent studies have demonstrated that modifying the 'classic' 6x30-s 'all-out' sprint
30 interval training (SIT) protocol by incorporating either shorter sprints (6x10-s or 15-s sprints) or
31 fewer sprints (e.g. 2x20-s sprints; reduced-exertion high-intensity interval training (REHIT)) does not
32 attenuate the training-induced improvements in maximal aerobic capacity ($\dot{V}O_2\text{max}$). The aim of the
33 present study was to determine whether reducing the sprint duration in the REHIT protocol from 20
34 s to 10 s per sprint influences acute affective responses and the change in $\dot{V}O_2\text{max}$ following training.

35 **Methods:** Thirty-six sedentary or recreationally active participants (17 women; mean $\pm$ SD age: 22 $\pm$ 3
36 y, BMI: 24.5 $\pm$ 4.6 kg $\cdot$ m⁻², $\dot{V}O_2\text{max}$: 37 $\pm$ 8 mL $\cdot$ kg⁻¹ $\cdot$ min⁻¹) were randomised to a group performing a
37 'standard' REHIT protocol involving 2x20-s sprints or a group who performed 2x10-s sprints. $\dot{V}O_2\text{max}$
38 was determined before and after 6 weeks of 3 weekly training sessions. Acute affective responses
39 and perceived exertion were assessed during training.

40 **Results:** Greater increases in $\dot{V}O_2\text{max}$ were observed for the group performing 20-s sprints
41 (2.77 $\pm$ 0.75 to 3.04 $\pm$ 0.75 L $\cdot$ min⁻¹; +10%) compared to the group performing 10-s sprints (2.58 $\pm$ 0.57 vs.
42 2.67 $\pm$ 3.04 L $\cdot$ min⁻¹; +4%; group $\times$ time interaction effect: $p < 0.05$; $d = 1.06$). Positive affect and the mood
43 state vigour increased post-exercise, while tension, depression and total mood disturbance
44 decreased, and negative affect remained unchanged. Affective responses and perceived exertion
45 were not altered by training and were not different between groups.

46 **Conclusion:** Reducing sprint duration in the REHIT protocol from 20 s to 10 s attenuates
47 improvements in $\dot{V}O_2\text{max}$, and does not result in more positive affective responses or lower
48 perceived exertion.

49

50

51 **Key words:**52 $\dot{V}O_2\text{max}$; sprint interval training; SIT; Wingate sprint; affect

53

54 **Abbreviations:**

55 ANOVA: analysis of variance; BMI: body mass index; BRUMS: Brunel mood scale; HIIT: high-intensity
56 interval training; IPAQ: International Physical Activity Questionnaire; MICT: moderate-intensity
57 continuous training; PANAS: positive and negative affect schedule; PAR-Q: physical activity readiness
58 questionnaire; RCT: randomised controlled trial; REHIT: reduced-exertion high-intensity interval
59 training; RPE: rating of perceived exertion; SIT: sprint interval training; TMD: total mood disturbance;
60 $\dot{V}O_2\text{max}$: maximal aerobic capacity

Draft

61 INTRODUCTION

62 Low maximal aerobic capacity ($\dot{V}O_{2\max}$) is one of the strongest predictors of future chronic disease
63 and premature mortality (Keteyian et al. 2008; Myers et al. 2002), and longitudinal studies have
64 demonstrated that increasing $\dot{V}O_{2\max}$ substantially lowers morbidity and mortality during follow up
65 (Blair et al. 1995; Lee et al. 2011). Thus physical activity and/or exercise interventions that improve
66 $\dot{V}O_{2\max}$ should be emphasised in public health guidelines (Bouchard et al. 2015; Ross et al. 2016).
67 $\dot{V}O_{2\max}$ can be improved through moderate-intensity continuous training (MICT) (Garber et al.
68 2011), but as perceived lack of time prevents many people from doing such exercise (Kimm et al.
69 2006; Korkiakangas et al. 2009; Trost et al. 2002), alternative exercise routines such as (sub)maximal
70 high-intensity interval training (HIIT) and supramaximal sprint interval training (SIT) have been
71 proposed as time-efficient alternative/adjunct exercise strategies (Gillen and Gibala 2014). However,
72 the need for recovery periods in between repeated sprints diminishes the time-efficiency of most
73 HIIT and SIT protocols compared to less strenuous MICT (Gillen and Gibala 2014).

74 The mechanisms which underpin the increases in $\dot{V}O_{2\max}$ with HIIT and SIT remain unclear (Vollaard
75 and Metcalfe 2017), but have been proposed to rely on activation of skeletal muscle signalling
76 pathways involving 5' adenosine monophosphate-activated protein kinase (AMPK), peroxisome
77 proliferator-activated receptor gamma coactivator 1-alpha (PGC-1 α) (Gibala 2009), and potentially
78 other signalling molecules such as p38 mitogen-activated protein kinase (p38 MAPK) (Little et al.
79 2011), sirtuin 1 (SIRT1) (Guerra et al. 2010) and sirtuin 3 (SIRT3) (Edgett et al. 2016). It has been
80 proposed that the activation of relevant signalling pathways with supramaximal exercise may be
81 reliant on achieving peak power output (Hazell et al. 2010), but we have proposed an alternative
82 theory suggesting that activation of signalling pathways could plausibly be related to rapid
83 glycogenolysis associated with supramaximal exercise (Metcalfe et al. 2015). As glycogen depletion
84 during supramaximal exercise is limited to the first ~15 s of the first bouts of repeated 'all-out'
85 sprints (Parolin et al. 1999), we have previously proposed that more time-efficient SIT protocols with
86 fewer and/or shorter sprints may remain effective at improving $\dot{V}O_{2\max}$ (Metcalfe et al. 2012;

87 Vollaard and Metcalfe 2017). In recent years we have provided support for this hypothesis by
88 demonstrating that a reduced-exertion high-intensity interval training (REHIT) protocol involving two
89 20-s all-out cycle sprints within a 10-min exercise session improves $\dot{V}O_2\text{max}$ in sedentary individuals
90 (Metcalfe et al. 2012; Metcalfe et al. 2016) and patients with type 2 diabetes (Ruffino et al. 2017).
91 Furthermore, REHIT may improve measures of insulin sensitivity (Gillen et al. 2016; Gillen et al. 2014;
92 Metcalfe et al. 2012) and blood pressure (Ruffino et al. 2017). With a total time-commitment of 3 x
93 10 min per week, the REHIT protocol appears to offer a genuinely time-efficient alternative to MICT
94 (Vollaard and Metcalfe 2017; Vollaard et al. 2017).

95 The sprints in the REHIT protocol are shorter (20 s) compared to those used in most other SIT
96 protocols (30 s). This shortened sprint duration, and the concomitant lower increase in muscle
97 metabolites (e.g. lactate, hydrogen ions) and reduction in central motor command can be expected
98 to reduce perceived exertion (Pageaux 2016; Vollaard and Metcalfe 2017). The strong contribution
99 of phosphocreatine hydrolysis to energy demands during the first ~10 s of a 30-s Wingate sprint
100 means that perceived exertion during this phase is relatively low, whereas the gradual switch to
101 glycolysis as the predominant energy source (Parolin et al. 1999) is associated with severe and
102 progressive fatigue during the latter stages of the sprint. It follows that if sprint duration in the REHIT
103 protocol can be reduced further without affecting the associated improvements in $\dot{V}O_2\text{max}$, the
104 protocol would be perceived as less unpleasant and more tolerable. Recent findings by Townsend et
105 al. (2017) suggest that reducing sprint duration may attenuate negative affective responses to SIT.
106 This in turn could potentially lead to better uptake of, and adherence to, SIT as a viable alternative
107 exercise intervention for reducing risk of noncommunicable disease (Rhodes and Kates 2015).

108 To date there has been little focus on the effects of sprint duration in SIT protocols on associated
109 adaptations to $\dot{V}O_2\text{max}$. Hazell and co-workers (2010) suggested that generation of peak power may
110 be more important as a mechanism for improving $\dot{V}O_2\text{max}$ than the maintenance of a high power
111 output for a longer duration. Indeed, they observed no difference in the increase in $\dot{V}O_2\text{max}$
112 between SIT protocols incorporating either 4-6 x 30-s or 10-s sprints (Hazell et al. 2010). This was

113 supported by Zelt et al. (2014) who found similar increases in $\dot{V}O_{2\max}$ with 4-6 x 30-s or 15-s sprints.
114 Therefore, the present study aimed to determine whether reducing the sprint duration in the REHIT
115 protocol from 20 s to 10 s per sprint affects the associated change in $\dot{V}O_{2\max}$ as well as ratings of
116 perceived exertion and changes in affect and mood state.

Draft

117 **METHODS**

118 **Compliance with Ethical Standards**

119 The study was approved by the local University Ethics committees (references: SSREC 888/ FC52
120 2016-17/ 16-8.1/18), and conformed to the standards set forth in the latest revision of the
121 Declaration of Helsinki. The study protocol was fully explained to all participants in written and
122 verbal form before they were asked to provide written consent.

123 **Participants**

124 Thirty-six apparently healthy, sedentary or recreationally active participants (**Table 1**) were recruited
125 at sites in the UK (Stirling, Derry/Londonderry) and Turkey (İzmir), and randomised into a group
126 performing the REHIT protocol as previously described by Metcalfe et al. (2012) (REHIT20; n=18, 11
127 men) and a group performing the same protocol but with the sprint duration reduced by 50% for
128 each session (REHIT10; n=18, 8 men). Randomisation was performed using the sealed envelope
129 method. Exclusion criteria were classification as highly physically active according to the
130 International Physical Activity Questionnaire (IPAQ (Craig et al. 2003)), contraindications to exercise
131 as determined using a standard physical activity readiness questionnaire (PAR-Q (Thomas et al.
132 1992)), clinically significant hypertension (>140/90 mm Hg), or resting heart rate ≥ 100 beats·min⁻¹.
133 All participants were asked not to make conscious changes to their diet and physical activity patterns
134 for the duration of the study.

135 **Experimental Procedures**

136 Following measurement of body mass and height, $\dot{V}O_2$ max was determined using an incremental
137 cycling test to exhaustion (Excalibur Sport, Lode, Groningen, the Netherlands), with breath-by-
138 breath measurement of oxygen uptake using a calibrated online gas analyser (Oxycon Pro, Jaeger,
139 Wurzburg, Germany; Quark C-PET, Cosmed, Rome, Italy; CV: 4%). Participants were asked not to
140 perform strenuous exercise or consume caffeine or alcohol the day before and prior to the test, and
141 to drink half a litre of water the morning of the testing day. After a 2-min warm-up at 50 W the

intensity was increased by 1 W every 3 s until volitional exhaustion (failure to maintain rpm >50) despite verbal encouragement. $\dot{V}O_2$ max was determined as the highest value for a 15-breath rolling average, and accepted if two or more of the following criteria were met: 1) volitional exhaustion, 2) a plateau in $\dot{V}O_2$ despite increasing intensity, 3) $RER > 1.15$, and 4) maximal heart rate within 10 beats of the age-predicted maximum (i.e. $220 - \text{age}$). This was the case for all participants. Because of technical difficulties we were unable to complete the post-training $\dot{V}O_2$ max test for one female participant in the REHIT20 group, so $\dot{V}O_2$ max data are presented for 11 men and 6 women for this group.

Following the $\dot{V}O_2$ max test, participants performed a 6-week training programme involving 3 training sessions per week. Training sessions consisted of 10 min of unloaded pedalling (Ergomedic 874e, Monark, Vansbro, Sweden), with two all-out cycle sprints against a resistance of 7.5% of the participant's body weight. The first sprint started at 2 min and the second sprint finished at 6 min. Sprint duration in the REHIT20 group increased from 10 s in the first 3 sessions, to 15 s in sessions 4-6, and 20 s in all remaining sessions. Sprint duration in the REHIT10 group was exactly half that of the REHIT20 group in all sessions: duration increased from 5 s in the first 3 sessions, to 7.5 s in sessions 4-6, and 10 s in all remaining sessions. Participants were asked to increase their cadence to maximal ~ 3 s prior to the start of the sprint and to maintain the highest possible cadence throughout the sprint. Verbal encouragement was provided. A session rating of perceived exertion (RPE) score using the 6-20 Borg scale (representing the entire training session (Borg 1982)) was taken directly post-exercise at the last session of each training week. In addition, during the first, fourth, seventh and eighteenth training session, the changes in psychological affect as a result of exercise were assessed using the positive and negative affect scale (PANAS (Watson et al. 1988)) and the Brunel Mood Scale (BRUMS (Terry et al. 1999)). These scales were completed before and within 5 min after the exercise session. Following the final training session, participants completed a questionnaire on the acceptability of the intervention, as previously used by Boereboom et al. (2016). A second

167 $\dot{V}O_2$ max test was performed on the third day after the final training session, at a similar time as the
168 baseline test and following identical procedures.

169 **Statistical analysis**

170 Data are presented as mean $\pm$ SD. Based on a coefficient of variation for the $\dot{V}O_2$ max test protocol of
171 4% (Songsorn et al. 2016), it was calculated that 14 participants were needed in each group to be
172 able to detect a difference in the change in $\dot{V}O_2$ max of 5% between the two groups, with a power of
173 90% and $\alpha=0.05$. Independent sample t-tests were used to determine differences between the two
174 groups at baseline. Two-way mixed model ANOVA (intervention [REHIT10 / REHIT20] $\times$ trial [pre-
175 training / post-training]) was used to determine differences in the change in $\dot{V}O_2$ max and body mass
176 from baseline to post-intervention between the two groups. Three-way repeated measures ANOVA
177 (intervention [REHIT10 / REHIT20] $\times$ training session [session 1 / 4 / 7 / 18] $\times$ time [pre-exercise /
178 post-exercise]) was used to assess the effect of acute exercise on affect and mood state, and two-
179 way repeated measures ANOVA (intervention [REHIT10 / REHIT20] $\times$ training session [session 3 / 6 /
180 9 / 12 / 15 / 18]) was used to assess the effect of acute exercise on perceived exertion. A Mann-
181 Whitney U test was used to test for differences between the groups for intervention acceptability.
182 Cohen's *d* effect sizes are reported. Significance was accepted at $p<0.05$.

RESULTS

There were no significant differences between participants in the control group and the training group for mean baseline characteristics (**Table 1**) or other variables. Of the 36 participants, 23 completed all 18 training sessions, 7 completed 17 sessions, and 6 completed 16 sessions, resulting in an overall mean adherence of 97% (98% for REHIT10 and 96% for REHIT20). We observed a small but significant increase in body mass from pre- to post-intervention (main effect of time: $p<0.05$), but no difference between the change in the REHIT20 group (70.3 ± 11.5 vs. 71.2 ± 11.0 kg) compared to the REHIT10 group (75.8 ± 18.7 vs. 76.2 ± 18.8 kg). There was a greater increase in $\dot{V}O_{2\max}$ in the REHIT20 group (2.77 ± 0.75 vs. 3.04 ± 0.75 L $\cdot$ min $^{-1}$; +10%) compared to the REHIT10 group (2.58 ± 0.57 vs. 2.67 ± 3.04 L $\cdot$ min $^{-1}$; +4%; group $\times$ time interaction effect: $P<0.05$; $d=1.06$; main effect of time: $p<0.001$; **Figure 1**).

Acute REHIT sessions were associated with a small but significant increase in positive affect directly post-exercise (main effect of time: $p<0.05$; $d=0.23$), whereas negative affect was unchanged (**Table 2**). There were no significant time $\times$ session interaction effects for affect, and no significant differences were observed between the changes in the REHIT10 and REHIT20 groups. Significant exercise-induced improvements with small effect sizes ($d=0.11$ - 0.21) were observed for the mood states tension, depression, vigour (main effect of time: $p<0.01$ for each) and total mood disturbance (main effect of time: $p<0.05$), but there were no significant changes for anger, fatigue, or confusion (**Figure 2**). Again, no differences were observed between training sessions, nor between the REHIT10 and REHIT20 groups. Ratings of perceived exertion scores averaged between ~ 13 - 15 ('somewhat hard' to 'hard'), with no significant differences between the groups or between training sessions (**Figure 3**). Following the final training session, both REHIT programmes were deemed acceptable by the majority of participants (**Figure 4**), with no significant differences between the protocols. Specifically, on average participants in both groups agreed with the statements that they believed that their fitness had improved, that they would do HIT again, that they would recommend HIT to others, and that they enjoyed HIT. They were neutral about the statement that HIT was more

209 demanding than expected, and on average disagreed with the statements that the physical strain or
210 travelling involved with HIT interfered with their life, and that HIT was a time burden.

Draft

211 DISCUSSION

212 The present study aimed to determine whether a REHIT protocol with 10-s sprints could be equally
213 effective at improving $\dot{V}O_2\text{max}$ compared to a protocol involving 20-s sprints. Unlike previous studies
214 that found no difference in the increase in $\dot{V}O_2\text{max}$ between 2-4 weeks of 4-6 x 30-s sprints vs. 10
215 (Hazell et al. 2010) or 15-s sprints (Zelt et al. 2014), we observed significantly greater improvements
216 in $\dot{V}O_2\text{max}$ with 2 x 20-s vs. 2 x 10-s sprints. As acute affective responses and perceived exertion
217 were not affected by sprint duration, we conclude that in protocols with very few sprints the sprint
218 duration should be longer than 10 s.

219 The increase in $\dot{V}O_2\text{max}$ we observed in response to our original 2x20-s REHIT protocol is similar to
220 what we have reported in our previous studies (Metcalfe et al. 2012; Metcalfe et al. 2016). This
221 increase is important as low $\dot{V}O_2\text{max}$ is strongly linked to poor health and increased risk of
222 premature morbidity (Keteyian et al. 2008; Myers et al. 2002). As the prevalence of physical
223 inactivity remains high worldwide (Hallal et al. 2012), there is an urgent need to establish alternative
224 exercise interventions that are 1) effective at improving $\dot{V}O_2\text{max}$, and 2) acceptable to populations
225 that are currently unwilling or unable to adhere to recommended levels of MICT. In an attempt to
226 address the commonly reported barrier to exercise of perceived lack of time (Kimm et al. 2006;
227 Korkiakangas et al. 2009; Trost et al. 2002) we previously developed the genuinely time-efficient
228 REHIT intervention (Metcalfe et al. 2012). REHIT has a total time-commitment of 3 x 10 min per
229 week and is associated with acceptable session ratings of perceived exertion ('somewhat hard')
230 (Metcalfe et al. 2012). Nonetheless, there remains a need to establish whether the REHIT protocol
231 can be made shorter and/or easier in order to enhance the likelihood of sedentary populations
232 taking up and adhering to this exercise routine. We have recently provided preliminary data
233 suggesting that a minimum of two sprint repetitions within a training session is required in order to
234 improve $\dot{V}O_2\text{max}$ (Songsorn et al. 2016; Songsorn et al. 2017; Vollaard et al. 2017). Reducing the
235 sprint duration in the REHIT protocol would do little to reduce the total training time (the bulk of
236 each training session is spent on the low-intensity warm-up and recovery periods), but if the

resulting protocol remains effective at improving $\dot{V}O_{2\max}$ it could be beneficial by reducing potential negative affective responses or perceived exertion (Townsend et al. 2017). However, in the present study we demonstrate that reducing the sprint duration from 20 s to 10 s attenuates the efficacy of the REHIT protocol at improving $\dot{V}O_{2\max}$, and we therefore propose that the original protocol involving 20-s sprints remains the protocol of choice.

The exact physiological stimuli and molecular signalling pathways responsible for the rapid improvements in $\dot{V}O_{2\max}$ associated with HIIT and SIT protocols remain unclear (Vollaard and Metcalfe 2017). As long as this is the case, establishing the shortest and/or easiest exercise interventions associated with health benefits such as improved $\dot{V}O_{2\max}$ remains a matter of trial and error, involving studies examining a variety of protocol modifications. Previous work by the groups of Hazell (Hazell et al. 2010) and Gurd (Zelt et al. 2014) suggested that the sprint duration in SIT protocols can be substantially reduced from that used in the 'classic' 4-6 x 30-s SIT protocol popularised by Gibala's group (Burgomaster et al. 2005). On the basis of these results it was suggested that generation of peak power is a key stimulus for improving $\dot{V}O_{2\max}$. Our present data suggest that this is not the case, at least not when the number of sprint repetitions is kept low (2 repetitions). The evidence to date points to a need to repeat all-out sprints of a sufficient duration (e.g. 2 x 20 s) or number (e.g. 4-6 x 10 s) to achieve clinically meaningful increases in $\dot{V}O_{2\max}$. The reasons why certain combinations of sprint repetitions and sprint duration lead to differences in the effectiveness of the protocols in improving $\dot{V}O_{2\max}$ will remain unclear until we have a more detailed understanding of the mechanisms responsible for the increase in $\dot{V}O_{2\max}$ associated with very low volume SIT protocols (Vollaard and Metcalfe 2017).

This is the first study to provide data on the acute effects of REHIT on affective responses and mood state. We demonstrate that reducing the sprint duration in the REHIT protocol from 20 s to 10 s does not reduce session ratings of perceived exertion, nor does it lead to more positive affective responses or changes in mood state post-exercise. This is important, because a criticism of SIT protocols has been that all-out sprints are too strenuous for untrained individuals: it has been

suggested that performing all-out sprints requires too much motivation (which many sedentary populations lack), and is likely to evoke negative affect which may lead to subsequent avoidance of further exercise (Hardcastle et al. 2014). However, the evidence on which this criticism is based is contentious (Jung et al. 2015), and our data suggest that REHIT is not associated with the negative affective responses described by Hardcastle et al. (2014).

Exercise interventions can only improve health if performed regularly, and therefore exercise protocols leading to a smaller increase in $\dot{V}O_2\text{max}$ could still be deemed acceptable if they were substantially less strenuous and associated with less negative affect. However, we demonstrate that acute exercise in a REHIT session is not associated with an increase in negative affect or a disruption of mood state directly after exercise, and that neither reducing the sprint duration from 20 s to 10 s nor regularly performing REHIT sessions (i.e. training) modifies the acute responses. Our finding that REHIT does not result in negative affect directly post-exercise is in contrast to longer SIT protocols with more sprints (Townsend et al. 2017). Conversely it is in line with Townsend's finding that shorter and, if sprint duration is kept constant, fewer sprints have less of an impact on affective responses. Nonetheless, our data should be interpreted with caution given that affective responses and changes in mood state were only assessed on completion of the exercise session, i.e. ~5 min following the second sprint. Previous studies which have applied continuous exercise have shown that affective responses to exercise above the ventilatory threshold follow an inverse U-shaped curve: affect decreases (and may become negative) during exercise, but then rebounds quickly upon exercise cessation and mood state may then be enhanced in the immediate post-exercise period (Ekkekakis et al. 2008). Future studies should determine whether negative affect is present at any point during the 10-min exercise session. Still, that REHIT may be associated with positive affective responses immediately post-exercise is in direct contrast to classic SIT (4-6 x 30-s sprints) (Saaniijoki et al. 2015) and at the very least suggests that any potential negative responses will be short-lived.

In conclusion, our present study shows that reducing the sprint duration in the REHIT protocol from 2 x 20 s to 2 x 10 s attenuates improvements in $\dot{V}O_2\text{max}$, but does not impact perceived exertion,

289 acute affective responses or mood state. Our findings further support the use of the original REHIT
290 protocol consisting of 2 x 20-s sprints, and strengthens our contention that REHIT represents an
291 efficacious, time-efficient, and acceptable alternative to classic SIT and MICT for improving health
292 and reducing risk of future morbidity and premature mortality.

Draft

293 **Acknowledgements**

294

295 We would like to thank Evangeline Nicol, Jeremy Basset Sagarminaga, Stefanos Strzemieski, Noel

296 McLaughlin, Patrick Meenan, and Sam Russell for assistance with testing and the training sessions.

297 PS is supported by a PhD studentship provided by Thammasat University.

298

299 **Conflict of interest**

300 The authors declare that they have no conflict of interest.

Draft

REFERENCES

- Blair SN, Kohl HW, 3rd, Barlow CE, Paffenbarger RS, Jr., Gibbons LW, Macera CA (1995) Changes in physical fitness and all-cause mortality. A prospective study of healthy and unhealthy men. *Jama* 273 (14):1093-1098
- Boereboom CL, Phillips BE, Williams JP, Lund JN (2016) A 31-day time to surgery compliant exercise training programme improves aerobic health in the elderly. *Tech Coloproctol* 20 (6):375-382. doi:10.1007/s10151-016-1455-1
- Borg GA (1982) Psychophysical bases of perceived exertion. *Medicine and science in sports and exercise* 14 (5):377-381
- Bouchard C, Blair SN, Katzmarzyk PT (2015) Less Sitting, More Physical Activity, or Higher Fitness? *Mayo Clinic proceedings* 90 (11):1533-1540. doi:10.1016/j.mayocp.2015.08.005
- Burgomaster KA, Hughes SC, Heigenhauser GJ, Bradwell SN, Gibala MJ (2005) Six sessions of sprint interval training increases muscle oxidative potential and cycle endurance capacity in humans. *Journal of applied physiology* 98 (6):1985-1990. doi:10.1152/japplphysiol.01095.2004
- Craig CL, Marshall AL, Sjostrom M, Bauman AE, Booth ML, Ainsworth BE, Pratt M, Ekelund U, Yngve A, Sallis JF, Oja P (2003) International physical activity questionnaire: 12-country reliability and validity. *Medicine and science in sports and exercise* 35 (8):1381-1395. doi:10.1249/01.MSS.0000078924.61453.FB
- Edgett BA, Bonafiglia JT, Baechler BL, Quadrilatero J, Gurd BJ (2016) The effect of acute and chronic sprint-interval training on LRP130, SIRT3, and PGC-1alpha expression in human skeletal muscle. *Physiol Rep* 4 (17). doi:10.14814/phy2.12879
- Ekkekakis P, Hall EE, Petruzzello SJ (2008) The relationship between exercise intensity and affective responses demystified: to crack the 40-year-old nut, replace the 40-year-old nutcracker! *Ann Behav Med* 35 (2):136-149. doi:10.1007/s12160-008-9025-z
- Garber CE, Blissmer B, Deschenes MR, Franklin BA, Lamonte MJ, Lee IM, Nieman DC, Swain DP, American College of Sports M (2011) American College of Sports Medicine position stand. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: guidance for prescribing exercise. *Medicine and science in sports and exercise* 43 (7):1334-1359. doi:10.1249/MSS.0b013e318213fefb
- Gibala M (2009) Molecular responses to high-intensity interval exercise. *Applied physiology, nutrition, and metabolism = Physiologie appliquee, nutrition et metabolisme* 34 (3):428-432. doi:10.1139/H09-046
- Gillen JB, Gibala MJ (2014) Is high-intensity interval training a time-efficient exercise strategy to improve health and fitness? *Applied physiology, nutrition, and metabolism = Physiologie appliquee, nutrition et metabolisme* 39 (3):409-412. doi:10.1139/apnm-2013-0187
- Gillen JB, Martin BJ, MacInnis MJ, Skelly LE, Tarnopolsky MA, Gibala MJ (2016) Twelve Weeks of Sprint Interval Training Improves Indices of Cardiometabolic Health Similar to Traditional Endurance Training despite a Five-Fold Lower Exercise Volume and Time Commitment. *PloS one* 11 (4):e0154075. doi:10.1371/journal.pone.0154075
- Gillen JB, Percival ME, Skelly LE, Martin BJ, Tan RB, Tarnopolsky MA, Gibala MJ (2014) Three minutes of all-out intermittent exercise per week increases skeletal muscle oxidative capacity and improves cardiometabolic health. *PloS one* 9 (11):e111489. doi:10.1371/journal.pone.0111489
- Guerra B, Guadalupe-Grau A, Fuentes T, Ponce-Gonzalez JG, Morales-Alamo D, Olmedillas H, Guillen-Salgado J, Santana A, Calbet JA (2010) SIRT1, AMP-activated protein kinase phosphorylation and downstream kinases in response to a single bout of sprint exercise: influence of glucose ingestion. *Eur J Appl Physiol* 109 (4):731-743. doi:10.1007/s00421-010-1413-y

- 351 Hallal PC, Andersen LB, Bull FC, Guthold R, Haskell W, Ekelund U, Lancet Physical Activity Series
352 Working G (2012) Global physical activity levels: surveillance progress, pitfalls, and
353 prospects. *Lancet* 380 (9838):247-257. doi:10.1016/S0140-6736(12)60646-1
- 354 Hardcastle SJ, Ray H, Beale L, Hagger MS (2014) Why sprint interval training is inappropriate for a
355 largely sedentary population. *Front Psychol* 5:1505. doi:10.3389/fpsyg.2014.01505
- 356 Hazell TJ, Macpherson RE, Gravelle BM, Lemon PW (2010) 10 or 30-s sprint interval training bouts
357 enhance both aerobic and anaerobic performance. *Eur J Appl Physiol* 110 (1):153-160.
358 doi:10.1007/s00421-010-1474-y
- 359 Jung ME, Little JP, Batterham AM (2015) Commentary: Why sprint interval training is inappropriate
360 for a largely sedentary population. *Front Psychol* 6:1999. doi:10.3389/fpsyg.2015.01999
- 361 Keteyian SJ, Brawner CA, Savage PD, Ehrman JK, Schairer J, Divine G, Aldred H, Ophaug K, Ades PA
362 (2008) Peak aerobic capacity predicts prognosis in patients with coronary heart disease.
363 *American heart journal* 156 (2):292-300. doi:10.1016/j.ahj.2008.03.017
- 364 Kimm SY, Glynn NW, McMahon RP, Voorhees CC, Striegel-Moore RH, Daniels SR (2006) Self-
365 perceived barriers to activity participation among sedentary adolescent girls. *Medicine and*
366 *science in sports and exercise* 38 (3):534-540. doi:10.1249/01.mss.0000189316.71784.dc
- 367 Korkiakangas EE, Alahuhta MA, Laitinen JH (2009) Barriers to regular exercise among adults at high
368 risk or diagnosed with type 2 diabetes: a systematic review. *Health promotion international*
369 24 (4):416-427. doi:10.1093/heapro/dap031
- 370 Lee DC, Sui X, Artero EG, Lee IM, Church TS, McAuley PA, Stanford FC, Kohl HW, 3rd, Blair SN (2011)
371 Long-term effects of changes in cardiorespiratory fitness and body mass index on all-cause
372 and cardiovascular disease mortality in men: the Aerobics Center Longitudinal Study.
373 *Circulation* 124 (23):2483-2490. doi:10.1161/CIRCULATIONAHA.111.038422
- 374 Little JP, Safdar A, Bishop D, Tarnopolsky MA, Gibala MJ (2011) An acute bout of high-intensity
375 interval training increases the nuclear abundance of PGC-1alpha and activates mitochondrial
376 biogenesis in human skeletal muscle. *Am J Physiol Regul Integr Comp Physiol* 300 (6):R1303-
377 1310. doi:10.1152/ajpregu.00538.2010
- 378 Metcalfe RS, Babraj JA, Fawcner SG, Volvaard NB (2012) Towards the minimal amount of exercise for
379 improving metabolic health: beneficial effects of reduced-exertion high-intensity interval
380 training. *Eur J Appl Physiol* 112 (7):2767-2775. doi:10.1007/s00421-011-2254-z
- 381 Metcalfe RS, Koumanov F, Ruffino JS, Stokes KA, Holman GD, Thompson D, Volvaard NB (2015)
382 Physiological and molecular responses to an acute bout of reduced-exertion high-intensity
383 interval training (REHIT). *Eur J Appl Physiol* 115 (11):2321-2334. doi:10.1007/s00421-015-
384 3217-6
- 385 Metcalfe RS, Tardif N, Thompson D, Volvaard NBJ (2016) Changes in aerobic capacity and glycaemic
386 control in response to reduced-exertion high-intensity interval training (REHIT) are not
387 different between sedentary men and women. *Eur J Appl Physiol* 115 (11):1117-1123
- 388 Myers J, Prakash M, Froelicher V, Do D, Partington S, Atwood JE (2002) Exercise capacity and
389 mortality among men referred for exercise testing. *The New England journal of medicine*
390 346 (11):793-801. doi:10.1056/NEJMoa011858
- 391 Pageaux B (2016) Perception of effort in Exercise Science: Definition, measurement and
392 perspectives. *European journal of sport science* 16 (8):885-894.
393 doi:10.1080/17461391.2016.1188992
- 394 Parolin ML, Chesley A, Matsos MP, Spriet LL, Jones NL, Heigenhauser GJ (1999) Regulation of skeletal
395 muscle glycogen phosphorylase and PDH during maximal intermittent exercise. *The*
396 *American journal of physiology* 277 (5 Pt 1):E890-900
- 397 Rhodes RE, Kates A (2015) Can the Affective Response to Exercise Predict Future Motives and
398 Physical Activity Behavior? A Systematic Review of Published Evidence. *Ann Behav Med* 49
399 (5):715-731. doi:10.1007/s12160-015-9704-5
- 400 Ross R, Blair SN, Arena R, Church TS, Despres JP, Franklin BA, Haskell WL, Kaminsky LA, Levine BD,
401 Lavie CJ, Myers J, Niebauer J, Sallis R, Sawada SS, Sui X, Wisloff U, American Heart

- Association Physical Activity Committee of the Council on L, Cardiometabolic H, Council on Clinical C, Council on E, Prevention, Council on C, Stroke N, Council on Functional G, Translational B, Stroke C (2016) Importance of Assessing Cardiorespiratory Fitness in Clinical Practice: A Case for Fitness as a Clinical Vital Sign: A Scientific Statement From the American Heart Association. *Circulation* 134 (24):e653-e699. doi:10.1161/CIR.0000000000000461
- Ruffino JS, Songsorn P, Haggett M, Edmonds D, Robinson AM, Thompson D, Vollaard NB (2017) A comparison of the health benefits of reduced-exertion high-intensity interval training (REHIT) and moderate-intensity walking in type 2 diabetes patients. *Applied physiology, nutrition, and metabolism = Physiologie appliquee, nutrition et metabolisme* 42 (2):202-208. doi:10.1139/apnm-2016-0497
- Saanijoki T, Nummenmaa L, Eskelinen JJ, Savolainen AM, Vahlberg T, Kalliokoski KK, Hannukainen JC (2015) Affective Responses to Repeated Sessions of High-Intensity Interval Training. *Medicine and science in sports and exercise* 47 (12):2604-2611. doi:10.1249/MSS.0000000000000721
- Songsorn P, Lambeth-Mansell A, Mair JL, Haggett M, Fitzpatrick BL, Ruffino J, Holliday A, Metcalfe RS, Vollaard NB (2016) Exercise training comprising of single 20-s cycle sprints does not provide a sufficient stimulus for improving maximal aerobic capacity in sedentary individuals. *Eur J Appl Physiol* 116 (8):1511-1517. doi:10.1007/s00421-016-3409-8
- Songsorn P, Ruffino J, Vollaard NB (2017) No effect of acute and chronic supramaximal exercise on circulating levels of the myokine SPARC. *European journal of sport science* 17 (4):447-452. doi:10.1080/17461391.2016.1266392
- Terry PC, Lane AM, Lane HJ, Keohane L (1999) Development and validation of a mood measure for adolescents. *J Sports Sci* 17 (11):861-872. doi:10.1080/026404199365425
- Thomas S, Reading J, Shephard RJ (1992) Revision of the Physical Activity Readiness Questionnaire (PAR-Q). *Canadian journal of sport sciences = Journal canadien des sciences du sport* 17 (4):338-345
- Townsend LK, Islam H, Dunn E, Eys M, Robertson-Wilson J, Hazell TJ (2017) Modified sprint interval training protocols. Part II. Psychological responses. *Applied physiology, nutrition, and metabolism = Physiologie appliquee, nutrition et metabolisme* 42 (4):347-353. doi:10.1139/apnm-2016-0479
- Trost SG, Owen N, Bauman AE, Sallis JF, Brown W (2002) Correlates of adults' participation in physical activity: review and update. *Medicine and science in sports and exercise* 34 (12):1996-2001. doi:10.1249/01.MSS.0000038974.76900.92
- Vollaard NB, Metcalfe RS (2017) Research into the Health Benefits of Sprint Interval Training Should Focus on Protocols with Fewer and Shorter Sprints. *Sports medicine* Epub: 8 April. doi:10.1007/s40279-017-0727-x
- Vollaard NBJ, Metcalfe RS, Williams S (2017) Effect of Number of Sprints in an SIT Session on Change in VO2max: A Meta-analysis. *Medicine and science in sports and exercise* 49 (6):1147-1156. doi:10.1249/MSS.0000000000001204
- Watson D, Clark LA, Tellegen A (1988) Development and validation of brief measures of positive and negative affect: the PANAS scales. *J Pers Soc Psychol* 54 (6):1063-1070
- Zelt JG, Hankinson PB, Foster WS, Williams CB, Reynolds J, Garneys E, Tschakovsky ME, Gurd BJ (2014) Reducing the volume of sprint interval training does not diminish maximal and submaximal performance gains in healthy men. *Eur J Appl Physiol* 114 (11):2427-2436. doi:10.1007/s00421-014-2960-4

448 **Table 1** participant characteristics

	REHIT10 (n=18)	REHIT20 (n=18)
Sex (male / female)	8 / 10	11 / 7
Age (y)	22±4	22±2
BMI (kg·m ⁻²)	25.3±5.9	23.8±24.1
Baseline $\dot{V}O_2$ max (mL·kg ⁻¹ ·min ⁻¹)	34±9	39±7
Physical activity level (MET·min·week ⁻¹)	911±739	981±769

449 *Values shown are means±SD. Physical activity level was determined using the IPAQ.*

Draft

450 **Table 2** Effects of REHIT10 and REHIT20 on positive and negative affect

	Positive affect				Negative affect			
	REHIT10		REHIT20		REHIT10		REHIT20	
	Pre-exercise	Post-exercise	Pre-exercise	Post-exercise	Pre-exercise	Post-exercise	Pre-exercise	Post-exercise
Session 1	24±7	26±7	22±7	22±6	15±4	16±6	14±3	13±4
Session 4	22±6	26±7	19±7	19±7	15±4	15±5	13±5	13±5
Session 7	23±6	25±6	19±8	20±10	15±5	15±5	13±3	13±4
Session 18	25±6	26±8	20±9	21±9	15±5	16±5	14±5	14±5

Draft

Figure 1 Changes in $\dot{V}O_2\text{max}$ in response to 6 weeks of REHIT10 and REHIT20. The mean 10% increase in $\dot{V}O_2\text{max}$ following REHIT20 was significantly higher than the 4% increase following REHIT10 ($p<0.05$).

Figure 2 Effects of REHIT10 and REHIT20 on mood state as determined using the Brunel Mood Scale (BRUMS). As no main or interaction effects of training session were observed, values for training sessions 1, 4, 7, and 18 were averaged. Values shown are means $\pm$ SD. TMD: total mood disturbance. Main effects of time: * $p<0.05$, ** $p<0.01$.

Figure 3 Rating of perceived exertion (RPE) as taken at the end of the last training session of each week. Participants were asked to rate the 10-min training sessions as a whole. No significant differences were observed between REHIT10 and REHIT20, nor between the training sessions.

Figure 4 Acceptability of REHIT. Acceptability was assessed using a modified questionnaire as used by Boereboom et al. (2016). Answers were given on a five-point Likert scale ranging from 1 - strongly agree to 5 - strongly disagree. No significant differences in the mean responses were observed between REHIT10 and REHIT20.

Draft







